

Cyril Shih-Huan Hsu

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[Website](#) / [Google Scholar](#) / [LinkedIn](#)

Professional Summary

AI researcher with a PhD in Informatics, specializing in deep reinforcement learning, Transformer-based architectures, generative models, and multi-agent systems. Research focuses on decision-making, optimization, meta-learning, and coordination in complex networked environments, with applications spanning large-scale systems and decentralized agentic systems. Published in top-tier IEEE Transactions and leading international conferences.

Education

PhD's degree in Informatics <i>Doctor of Science</i>	<i>University of Amsterdam</i> 2021 - 2025
Master's degree in Electrical Engineering <i>Master of Science</i>	<i>National Taiwan University</i> 2013 - 2015
Bachelor's degree in Bio-environmental Systems Engineering <i>Bachelor of Science</i>	<i>National Taiwan University</i> 2009 - 2013

Work experience

University of Amsterdam <i>Postdoctoral Researcher</i>	2025 - present <i>Amsterdam, The Netherlands</i>
• Agentic AI-Driven Network Automation	
University of Amsterdam <i>Teaching Assistant/Student Thesis Supervisor</i>	2021 - 2025 <i>Amsterdam, The Netherlands</i>
• Advanced Deep Learning (Course) • Advanced Networking (Course) • Resource Optimization for Facial Recognition Systems (Thesis) • Multi-domain Simulation Framework for Virtualized Infrastructures (Thesis)	
NEC Laboratories Europe <i>Visiting Researcher</i>	2024 - 2024 <i>Heidelberg, Germany</i>
• Vision Transformer-Based Framework for Real-Time Radio Quality Prediction • Distributed Vision-Language Model-Based Multi-Robot Coordination System	
Deep Cube SA <i>CTO & Co-founder</i>	2018 - 2021 <i>Lausanne, Switzerland</i>
• Glaucoma Screening System • Skin Disorder Analysis Device • Data Collection Platform for Medical Centers	
Relajet Technologies Ltd. <i>Technical Director of Artificial Intelligence</i>	2018 - 2020 <i>Taipei, Taiwan</i>
• Real-time Voice Identification/Speech Enhancement System	

- Real-time Facial Recognition System
- Multimodal Interactive Chatbot
- Deep Neural Networks for Embedded System

Language proficiencies

English	Fluent
Mandarin	Native

Selected Publications

Conference	Hsu, C. S. H., & Yu, T. L. (2015). Optimization by Pairwise Linkage Detection, Incremental Linkage Set, and Restricted/Back mixing: DSMGA-II. In Proceedings of the 2015 Annual Conference on Genetic and Evolutionary Computation (GECCO).
Conference	Chang, W. Y., Hsu, C. S. H., & Chien, J. H. (2017). FATAUVA-Net: An Integrated Deep Learning Framework for Facial Attribute Recognition, Action Unit Detection, and Valence-Arousal Estimation. In 2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW).
Conference	Sun, M. C., Hsu, C. S. H., Yang, M. C., & Chien, J. H. (2018). Context-Aware Cascade Attention-Based RNN for Video Emotion Recognition. In 2018 First Asian Conference on Affective Computing and Intelligent Interaction (ACII Asia).
Conference	Hsu, C. S. H., Martín-Pérez, J., Papagianni, C., & Grosso, P. (2023). V2N Service Scaling with Deep Reinforcement Learning. In 2023 IEEE Conference on Network Operations and Management Symposium (NOMS).
Conference	Hsu, C. S. H., De Vleeschauwer, D., Papagianni, C., & Grosso, P. (2025). Online SLA Decomposition: Enabling Real-Time Adaptation to Evolving Network Systems. In 2025 Joint European Conference on Networks and Communications & 6G Summit (EuCNC).
Conference	Hsu, C. S. H., Papagianni, C., & Grosso, P. (2025). RAILS: Risk-Aware Iterated Local Search for Joint SLA Decomposition and Service Provider Management in Multi-Domain Networks. In 2025 IEEE Conference on High Performance Switching and Routing (HPSR).
Conference	Hsu, C. S. H., Li, X., Zanzi, L., Yang, Z., Papagianni, C., & Costa-Perez, X. (2025). MapViT: A Two-Stage ViT-Based Framework for Real-Time Radio Quality Map Prediction in Dynamic Environments. Submitted to the 2026 IEEE International Conference on Communications (ICC).
Conference	Dalgkitis, A., Hsu, C. S. H., Papagianni, C., Grosso, P., & de Laat, C. (2025). LLM-based Optimization Algorithm Selection for High-Performance Networks Orchestration. Proceedings of the SC '25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC Workshops).
Journal	Hsu, C. S. H., De Vleeschauwer, D., & Papagianni, C. (2023). SLA Decomposition for Network Slicing: A Deep Neural Network Approach. In 2023 IEEE Networking Letters (NL).
Journal	Hsu, C. S. H., Martín-Pérez, J., De Vleeschauwer, D., Valcarengi, L., Li, X., & Papagianni, C. (2025). A Deep RL Approach on Task Placement and Scaling of Edge Resources for Cellular Vehicle-to-Network Service Provisioning. In 2025 IEEE Transactions on Network and Service Management (TNSM).
Journal	Hsu, C. S. H., Dalgkitis, A., Papagianni, C., & Grosso, P. (2025). Transformer-Empowered Actor-Critic Reinforcement Learning for Sequence-Aware Service Function Chain Partitioning. In 2026 IEEE Transactions on Network Science and Engineering (TNSE).